Lecture 18

Heisenberg group and the Weyl relations

Objectives. *The Heisenberg group is the simplest noncommutative nilpotent Lie group. This lecture builds its Schrödinger representation on $L^2(\mathbb{R})$ from translations and modulations, and proves the Weyl relations, which describe how translations and modulations commute up to a phase. We then state the Stone–von Neumann theorem. It says that a fixed nonzero central character, which records the scalars by which the centre acts, determines one irreducible unitary representation up to unitary equivalence. For central character zero, the irreducibles are one-dimensional characters, and the translation representation on $L^2(\mathbb{R})$ contains none of them.*

By the end of this lecture you will be able to:

- Compute products, inverses, commutators, the exponential map and Ad in the Heisenberg group.
- Prove that the Schrödinger formula is a strongly continuous unitary representation satisfying the Weyl relations, and compute its derivative on $C_c^\infty(\mathbb{R})$.
- State the Stone–von Neumann theorem with its hypotheses, and turn a Weyl pair into a representation to which it applies.
- Compute a Gaussian matrix coefficient and use it to contrast nonzero and zero central character.

18.1 The group law

Definition 18.1. The *Heisenberg group* H is the set of real matrices

$$(a,b,c) := \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a,b,c \in \mathbb{R},$$

with matrix multiplication.

Multiplying the matrices gives

$$(a,b,c)(a',b',c') = (a+a',\ b+b',\ c+c'+ab'). \tag{18.1}$$

The identity is $(0,0,0)$, and

$$(a,b,c)^{-1} = (-a,-b,-c+ab).$$

Indeed, the third coordinate of $(a,b,c)(-a,-b,-c+ab)$ is $c-c+ab-ab=0$. The group H is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$, hence a Lie group by the closed-subgroup theorem of Lecture 3. The coordinates identify H with $\mathbb{R}^3$, so H is connected and simply connected.

Swapping the factors in the group law (18.1) changes only the last coordinate. For $g=(a,b,c)$ and $g'=(a',b',c')$,

$$gg' = (0,0,ab'-a'b)\,g'g. \tag{18.2}$$

By the group law (18.1), each $(0,0,d)$ commutes with every element. So the commutation formula (18.2) shows that g is central exactly when $ab'-a'b=0$ for all a',b', that is, when $a=b=0$. The centre is therefore

$$Z(H)=\{(0,0,c)\,:\,c\in\mathbb{R}\}\cong\mathbb{R},$$

and $(a,b,c)Z(H)\mapsto(a,b)$ identifies $H/Z(H)$ with $\mathbb{R}^2$.

The Lie algebra $\mathfrak{h}$ has basis $X=E_{12}$, $Y=E_{23}$, $Z=E_{13}$, where E_{ij} are matrix units. Then

$$[X,Y]=Z,\qquad [X,Z]=[Y,Z]=0.$$

Let $W=xX+yY+zZ$. Multiplying matrix units gives $W^2=xyZ$ and $W^3=0$. So the exponential series of W stops after the quadratic term:

$$\exp(xX+yY+zZ)=I+W+\tfrac12 W^2=\big(x,\ y,\ z+\tfrac12 xy\big). \tag{18.3}$$

So $\exp\colon\mathfrak{h}\to H$ is a diffeomorphism with inverse $(a,b,c)\mapsto aX+bY+(c-\frac12 ab)Z$. In particular $\exp(sX)=(s,0,0)$, $\exp(sY)=(0,s,0)$ and $\exp(sZ)=(0,0,s)$.

For the adjoint action we show one representative step. By the group law (18.1),

$$g\exp(sX)g^{-1}=(a+s,b,c)(-a,-b,-c+ab)=(s,0,-sb)=\exp\big(s(X-bZ)\big).$$

Thus $\mathrm{Ad}(g)X=X-bZ$. In the same way $\mathrm{Ad}(g)Y=Y+aZ$ and $\mathrm{Ad}(g)Z=Z$. In the ordered basis (X,Y,Z),

$$D_{\mathrm{Ad}}(g)=\begin{pmatrix}1&0&0\\0&1&0\\-b&a&1\end{pmatrix},\qquad \det D_{\mathrm{Ad}}(g)=1.$$

With the convention $\Delta(g)=|\det\mathrm{Ad}(g)|^{-1}$ of Lecture 16, H is unimodular. Directly, left and right translations are polynomial maps of $\mathbb{R}^3$ with triangular Jacobian matrices and ones on the diagonal. So Lebesgue measure $da\,db\,dc$ is a left and right Haar measure.

18.2 The Schrödinger representation

We use $L^2(\mathbb{R})$ with $\langle f,h\rangle=\int_{\mathbb{R}}\overline{f(t)}\,h(t)\,dt$. We use the Lebesgue-integration inputs from Lecture 16: completeness of L^2, Fubini–Tonelli, dominated convergence and change of variables. In particular, dominated convergence justifies differentiation under the integrals below. We also use two standard facts. First, Lebesgue integrals are invariant under $t\mapsto t+a$. Second, the space $C_c(\mathbb{R})$ of continuous compactly supported functions, and its subspace $C_c^\infty(\mathbb{R})$ of smooth ones, are dense in $L^2(\mathbb{R})$.

Definition 18.2. Fix $\lambda \in \mathbb{R}$. For $g = (a, b, c) \in H$ and $f \in L^2(\mathbb{R})$, the *Schrödinger operator* $\Pi_\lambda(g)$ is defined by

$$(\Pi_\lambda(a,b,c)f)(t) = e^{i\lambda(c+bt)} f(t+a). \tag{18.4}$$

The *translations* $U(a)$ and *modulations* $V_\lambda(b)$ are defined by

$$(U(a)f)(t) = f(t+a), \qquad (V_\lambda(b)f)(t) = e^{i\lambda bt} f(t).$$

By the definitions, the Schrödinger operators factor as

$$\Pi_\lambda(a,b,c) = e^{i\lambda c}\, V_\lambda(b)\, U(a).$$

This matches the factorisation of group elements

$$(a,b,c) = (0,0,c)(0,b,0)(a,0,0).$$

Our $U(a)$ is the translation by $-a$ in the convention of Lecture 15.

Proposition 18.3. *For every $\lambda \in \mathbb{R}$, Π_λ is a strongly continuous unitary representation of H on $L^2(\mathbb{R})$, and $\Pi_\lambda(0,0,c) = e^{i\lambda c} I$. The* Weyl relations *hold:*

$$U(a)V_\lambda(b) = e^{i\lambda ab} V_\lambda(b)U(a), \qquad a, b \in \mathbb{R}. \tag{18.5}$$

Proof. Homomorphism. Let $g = (a,b,c)$ and $g' = (a',b',c')$. Applying the defining formula (18.4) twice,

$$\begin{aligned}(\Pi_\lambda(g)\Pi_\lambda(g')f)(t) &= e^{i\lambda(c+bt)}(\Pi_\lambda(g')f)(t+a)\\ &= e^{i\lambda(c+bt)} e^{i\lambda(c'+b'(t+a))} f(t+a+a')\\ &= e^{i\lambda(c+c'+ab'+(b+b')t)} f(t+a+a').\end{aligned}$$

By the group law (18.1), this is $(\Pi_\lambda(gg')f)(t)$. Also $\Pi_\lambda(0,0,0) = I$.

Unitarity. Since $|e^{i\lambda(c+bt)}| = 1$, translation invariance gives $\|\Pi_\lambda(g)f\|^2 = \int |f(t+a)|^2 dt = \|f\|^2$. Each $\Pi_\lambda(g)$ is an isometry with inverse $\Pi_\lambda(g^{-1})$, hence unitary.

Centre and Weyl relations. The first claim is the defining formula (18.4) with $a = b = 0$. For the second,

$$(U(a)V_\lambda(b)f)(t) = (V_\lambda(b)f)(t+a) = e^{i\lambda b(t+a)} f(t+a) = e^{i\lambda ab}(V_\lambda(b)U(a)f)(t).$$

Strong continuity. This is the only analytic step. First let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$, and let $|a| \le 1$. Then $\Pi_\lambda(g)f - f$ vanishes outside $[-R-1, R+1]$. On that interval, $|e^{i\theta} - 1| \le |\theta|$ gives

$$|(\Pi_\lambda(g)f)(t) - f(t)| \le |\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|),$$

where $\omega_f(\delta) = \sup\{|f(t+s) - f(t)| : t \in \mathbb{R},\ |s| \le \delta\}$. Uniform continuity of f gives $\omega_f(\delta) \to 0$ as $\delta \to 0$. Integrating the square over an interval of length $2R + 2$,

$$\|\Pi_\lambda(g)f - f\| \le (2R+2)^{1/2}\Big(|\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|)\Big) \longrightarrow 0 \quad (g \to e).$$

Now let $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$. Choose $h \in C_c(\mathbb{R})$ with $\|f - h\| < \varepsilon$. Since $\Pi_\lambda(g)$ is unitary,

$$\|\Pi_\lambda(g)f - f\| \le \|\Pi_\lambda(g)(f-h)\| + \|\Pi_\lambda(g)h - h\| + \|h - f\| < 2\varepsilon + \|\Pi_\lambda(g)h - h\|.$$

So $\limsup_{g \to e} \|\Pi_\lambda(g)f - f\| \le 2\varepsilon$ for every ε. Finally, $\|\Pi_\lambda(g)f - \Pi_\lambda(g_0)f\| = \|\Pi_\lambda(g_0^{-1}g)f - f\|$, and $g_0^{-1}g \to e$ as $g \to g_0$. ∎

Combining the commutation formula (18.2) with the scalar action of the centre in Proposition 18.3 gives

$$\Pi_\lambda(g)\Pi_\lambda(g') = e^{i\lambda(ab'-a'b)}\,\Pi_\lambda(g')\Pi_\lambda(g).$$

For $\lambda \neq 0$ the operators need not commute. For $\lambda = 0$ they commute.

Remark 18.4. Let $A = \{(0,b,c)\}$, a closed abelian normal subgroup, with character $\chi_\lambda(0,b,c) = e^{i\lambda c}$. Exercise 18.5 shows that the representation unitarily induced from χ_λ, as in Lecture 17, is equivalent to Π_λ.

Next we differentiate. Write $\mathcal{D} = C_c^\infty(\mathbb{R})$ and tf for the function $t \mapsto tf(t)$.

Proposition 18.5. *Each $\Pi_\lambda(g)$ maps $\mathcal{D}$ into $\mathcal{D}$. For $f \in \mathcal{D}$ and $W \in \{X, Y, Z\}$, the limit*

$$d\Pi_\lambda(W)f = \lim_{s\to 0} s^{-1}(\Pi_\lambda(\exp sW)f - f)$$

exists in $L^2(\mathbb{R})$, and

$$d\Pi_\lambda(X)f = f', \qquad d\Pi_\lambda(Y)f = i\lambda t f, \qquad d\Pi_\lambda(Z)f = i\lambda f.$$

On $\mathcal{D}$ we have $[d\Pi_\lambda(X), d\Pi_\lambda(Y)] = d\Pi_\lambda(Z) = d\Pi_\lambda([X,Y])$.

Proof. Translation and multiplication by the smooth function $e^{i\lambda(c+bt)}$ preserve smoothness and compact support. Let f vanish outside $[-R,R]$ and let $0 < |s| \le 1$. For X, Taylor's theorem gives

$$\left|\frac{f(t+s)-f(t)}{s} - f'(t)\right| \le \tfrac{1}{2}|s|\,\|f''\|_\infty,$$

and the left side vanishes outside $[-R-1, R+1]$. Its L^2 norm is at most $(2R+2)^{1/2}\tfrac{1}{2}|s|\|f''\|_\infty \to 0$. For Y, the bound $|e^{i\theta} - 1 - i\theta| \le \theta^2/2$ gives

$$\left|\frac{e^{i\lambda st}-1}{s} - i\lambda t\right| |f(t)| \le \tfrac{1}{2}\lambda^2|s|R^2|f(t)|,$$

whose L^2 norm tends to 0. For Z, $(e^{i\lambda s}-1)/s \to i\lambda$. Finally,

$$[d\Pi_\lambda(X), d\Pi_\lambda(Y)]f = (i\lambda t f)' - i\lambda t f' = i\lambda f = d\Pi_\lambda(Z)f. \qquad \blacksquare$$

Remark 18.6. Since Π_λ is strongly continuous (Proposition 18.3), $s \mapsto \Pi_\lambda(\exp sW)$ is a strongly continuous one-parameter unitary group. Its generator B_W, as defined in Lecture 15, has domain exactly the vectors f at which $s \mapsto \Pi_\lambda(\exp sW)f$ is differentiable at $s = 0$ in $L^2(\mathbb{R})$. Stone's theorem (Lecture 15) shows that B_W is skew-adjoint. So the physical generator $A_W = iB_W$ is self-adjoint and satisfies $\Pi_\lambda(\exp sW) = \exp(-isA_W)$. The derivative formulas of Proposition 18.5 show that $\mathcal{D}$ lies in the domain of A_W and that on $\mathcal{D}$

$$A_X f = if', \qquad A_Y f = -\lambda t f, \qquad A_Z f = -\lambda f.$$

In quantum mechanics a particle on a line has state space $L^2(\mathbb{R})$. Take $\lambda = 1/\hbar$, where $\hbar > 0$ is the reduced Planck constant. Put $P = -\hbar A_X$ (momentum) and $Q = -\hbar A_Y$ (position). On $\mathcal{D}$, $Qf = tf$, $Pf = -i\hbar f'$ and $[Q,P] = i\hbar$.

In terms of these operators, $U(a) = \exp(iaP/\hbar)$ and $V_\lambda(b) = \exp(ibQ/\hbar)$. So the Weyl relations (18.5) are the bounded, exponentiated form of the commutation relation.

We never use the commutation relation as a hypothesis. We state without proof that a commutation relation of unbounded operators on a dense domain does not by itself imply the Weyl relations. Exercise 18.3 shows that no bounded operators on a nonzero Hilbert space satisfy $[Q,P] = i\hbar I$.

18.3 The Stone–von Neumann theorem

Definition 18.7. A unitary representation ρ of H on a Hilbert space $\mathcal{H}$ has *central character* $\lambda \in \mathbb{R}$ if $\rho(0,0,c) = e^{i\lambda c} I$ for all $c \in \mathbb{R}$.

By the centre statement in Proposition 18.3, the representation Π_λ has central character λ. Now let ρ be an irreducible unitary representation of H. Schur's lemma for unitary representations (Kirillov, *Lectures on the Orbit Method*, 2004, Appendix V) makes each central operator $\rho(0,0,c)$ a scalar. These scalars form a continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$, hence have the form $e^{i\lambda c}$. So every irreducible unitary representation has a central character. Reducible ones need not, for example $\Pi_1 \oplus \Pi_2$.

Lemma 18.8. *Let σ and τ be strongly continuous one-parameter unitary groups on a Hilbert space $\mathcal{H}$, and let $\lambda \in \mathbb{R}$. Suppose $\sigma(a)\tau(b) = e^{i\lambda ab}\tau(b)\sigma(a)$ for all $a, b \in \mathbb{R}$. Then $\rho(a,b,c) = e^{i\lambda c}\tau(b)\sigma(a)$ defines a strongly continuous unitary representation ρ of H on $\mathcal{H}$ with central character λ.*

Proof. Move $\sigma(a)$ past $\tau(b')$ using the hypothesis:

$$\rho(g)\rho(g') = e^{i\lambda(c+c')}\tau(b)\sigma(a)\tau(b')\sigma(a') = e^{i\lambda(c+c'+ab')}\tau(b+b')\sigma(a+a') = \rho(gg').$$

Each $\rho(g)$ is unitary, and $\rho(0,0,c) = e^{i\lambda c}I$. For continuity at (a_0,b_0,c_0), let $f \in \mathcal{H}$ and put $h = \sigma(a_0)f$. Then

$$\|\tau(b)\sigma(a)f - \tau(b_0)\sigma(a_0)f\| \le \|\sigma(a)f - \sigma(a_0)f\| + \|\tau(b)h - \tau(b_0)h\| \to 0,$$

and the scalar factor $e^{i\lambda c}$ is continuous. ∎

We call a pair (σ,τ) satisfying the hypotheses of Lemma 18.8 a *Weyl pair* with parameter λ. By the Weyl relations (18.5) and the strong continuity in Proposition 18.3, the translations and modulations form a Weyl pair (U,V_λ) with parameter λ. Two Weyl pairs are *equivalent* when a single unitary operator from the first space to the second conjugates each group of the first pair into the corresponding group of the second.

Theorem 18.9 (Stone–von Neumann). *Fix $\lambda \neq 0$.*

(a) The Schrödinger representation Π_λ is irreducible.

(b) Let ρ be a strongly continuous unitary representation of H on a nonzero separable Hilbert space $\mathcal{H}$ with central character λ. Then $\mathcal{H}$ is a Hilbert direct sum of mutually orthogonal closed ρ-invariant subspaces. On each summand ρ is unitarily equivalent to Π_λ.

We cite the Stone–von Neumann theorem, Theorem 18.9, from Kirillov, *Lectures on the Orbit Method*, Chapter 2.

The Hilbert space of an irreducible representation of H is automatically separable. By irreducibility, every nonzero vector is cyclic. By strong continuity, the closed span of the orbit of such a vector is already generated by its orbit under a countable dense subset of H. This closed span is the whole space, and it is separable.

The Stone–von Neumann theorem, Theorem 18.9, has the following consequences.

- Every irreducible unitary representation of H with central character $\lambda \neq 0$ is unitarily equivalent to Π_λ.
- By the lemma on Weyl pairs, Lemma 18.8, every Weyl pair with parameter $\lambda \neq 0$ on a nonzero separable space is equivalent to a direct sum of copies of (U, V_λ). In particular, every Weyl pair with parameter $1/\hbar$ on a nonzero separable space is unitarily equivalent to a direct sum of copies of $(\exp(iaP/\hbar), \exp(ibQ/\hbar))$.
- Distinct values of λ give inequivalent representations Π_λ (Exercise 18.6(c) on inequivalence).

A standard proof, going back to von Neumann, turns on one step. Let ρ be a representation as in the uniqueness statement (b) of Theorem 18.9. For $f, h \in \mathcal{H}$, consider the sesquilinear form

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \left\langle f, \rho(\exp(aX+bY))h \right\rangle \, da\, db.$$

The Gaussian weight has integral $4\pi/|\lambda|$, so $|B(f,h)| \leq 2\|f\|\|h\|$. Hence the Riesz representation theorem for bounded sesquilinear forms gives a bounded operator E on $\mathcal{H}$ with $\|E\| \leq 2$ and $B(f,h) = \langle f, Eh\rangle$ for all $f, h \in \mathcal{H}$.

That proof shows, from the Weyl relations and strong continuity, that E is a nonzero orthogonal projection. Each unit vector $v \in E\mathcal{H}$ then generates one copy of Π_λ. In that copy, v corresponds to the Gaussian of the next section, § 18.4. The weight is that Gaussian's matrix coefficient, given by (18.6) at $g = \exp(aX + bY)$.

18.4 Gaussian matrix coefficients and central character zero

Let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$. Since $\int e^{-t^2} dt = \sqrt{\pi}$, $\|\varphi\| = 1$. We compute $\langle \varphi, \Pi_\lambda(g)\varphi\rangle$ for $g = (a,b,c)$. We use the calculus fact

$$\int_{\mathbb{R}} e^{-(t+w)^2} dt = \sqrt{\pi} \qquad (w \in \mathbb{C}).$$

For real w this is a translation. Write $w = u + iv$. The v-derivative of the integrand is $i\,\frac{d}{dt} e^{-(t+w)^2}$, which integrates to 0. Differentiation under the integral is justified on bounded v-intervals by an integrable Gaussian bound, including the derivative. So the integral does not depend on v.

By the defining formula (18.4), and since φ is real,

$$\langle \varphi, \Pi_\lambda(g)\varphi\rangle = \pi^{-1/2} e^{i\lambda c} \int_{\mathbb{R}} e^{-t^2/2 - (t+a)^2/2 + i\lambda b t}\, dt.$$

Complete the square with $w = \tfrac{1}{2}(a - i\lambda b)$:

$$-\tfrac{t^2}{2} - \tfrac{(t+a)^2}{2} + i\lambda b t = -t^2 - (a - i\lambda b)t - \tfrac{a^2}{2} = -(t+w)^2 + w^2 - \tfrac{a^2}{2},$$

$$w^2 - \tfrac{a^2}{2} = \tfrac{1}{4}\big(a^2 - 2i\lambda ab - \lambda^2 b^2\big) - \tfrac{a^2}{2} = -\tfrac{1}{4}\big(a^2 + \lambda^2 b^2\big) - \tfrac{i\lambda ab}{2}.$$

The integral is $\sqrt{\pi}$ times $e^{w^2 - a^2/2}$. Therefore

$$\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle = e^{i\lambda(c - ab/2)}\, e^{-(a^2+\lambda^2 b^2)/4}. \tag{18.6}$$

We make two checks. First, at $g = e$ the value is $1 = \|\varphi\|^2$. Second, replacing g by $g^{-1} = (-a, -b, -c + ab)$ conjugates the value, as unitarity requires.

By the exponential formula (18.3), $c - ab/2$ is the Z-coordinate of $\log g$, where log is the inverse of the exponential map. Hence at $\exp(xX + yY + zZ)$ the coefficient is $e^{i\lambda z} e^{-(x^2+\lambda^2 y^2)/4}$.

The modulus depends only on the image (a, b) in $H/Z(H) \cong \mathbb{R}^2$. For $\lambda \neq 0$ it decays in every direction, and

$$\int_{\mathbb{R}^2} \big|\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle\big|^2 da\, db = \int e^{-a^2/2} da \int e^{-\lambda^2 b^2/2} db = \sqrt{2\pi} \cdot \frac{\sqrt{2\pi}}{|\lambda|} = \frac{2\pi}{|\lambda|}.$$

Its reciprocal $|\lambda|/(2\pi)$ is the normalising constant in the form B above.

The operator E defined after the Stone–von Neumann theorem can be computed explicitly in this model. This computation verifies its normalisation. It is not a proof of the general theorem. Write E_λ for that operator with $\rho = \Pi_\lambda$. Since

$$\Pi_\lambda(\exp(aX + bY))f(t) = e^{i\lambda(bt + ab/2)} f(t + a),$$

we integrate first over b. For $f, h \in C_c(\mathbb{R})$, Cauchy–Schwarz in t and translation invariance give

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\, |f(t+a)|\, dt\, da\, db \le \frac{4\pi}{|\lambda|} \|f\| \|h\|.$$

Thus the Fubini–Tonelli theorem identifies the following kernel calculation with the form B defining E_λ. The Gaussian Fourier integral, obtained by the same square completion as above, gives

$$\int_{\mathbb{R}} e^{-\lambda^2 b^2/4} e^{i\lambda b(t + a/2)}\, db = \frac{2\sqrt{\pi}}{|\lambda|} e^{-(t+a/2)^2}.$$

Consequently, for $f \in C_c(\mathbb{R})$,

$$(E_\lambda f)(t) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2/4 - (t+a/2)^2} f(t + a)\, da.$$

Set $u = t + a$. The exponent simplifies to

$$-\frac{(u - t)^2}{4} - \frac{(t + u)^2}{4} = -\frac{t^2 + u^2}{2}.$$

We therefore obtain

$$(E_\lambda f)(t) = \varphi(t) \int_{\mathbb{R}} \overline{\varphi(u)} f(u)\, du = \varphi(t) \langle \varphi, f\rangle.$$

Both operators are bounded, so density extends the identity to all $L^2(\mathbb{R})$. Thus E_λ is exactly the orthogonal projection onto the Gaussian line. For a general representation ρ with central

character $\lambda \neq 0$, proving that the operator E is a nonzero projection is the key step of the standard proof. Here its rank-one integral kernel is visible.

Now set $\lambda = 0$. The defining formula (18.4) becomes pure translation, $(\Pi_0(a,b,c)f)(t) = f(t+a)$. The modulations become trivial, and the Gaussian coefficient (18.6) becomes $e^{-a^2/4}$. This does not decay in b, and the integral above diverges.

The irreducible unitary representations of H with central character 0 factor through $H/Z(H) \cong \mathbb{R}^2$. By Schur's lemma (Kirillov, Appendix V), irreducible unitary representations of an abelian group are one-dimensional. Hence the irreducible unitary representations of H with central character 0 are the characters

$$\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a+\beta b)}, \qquad \alpha,\beta \in \mathbb{R}.$$

Kirillov's Chapter 2 records this together with the Stone–von Neumann theorem, Theorem 18.9, as the full list of irreducibles. Lecture 19 explains the list through coadjoint orbits.

Proposition 18.10. *Π_0 has no one-dimensional invariant subspace. Hence it has no irreducible closed invariant subspace.*

Proof. Suppose $f \neq 0$ spans an invariant line. Then $U(a)f = \mu(a)f$ with $|\mu(a)| = 1$. For each integer n,

$$\int_n^{n+1} |f(t)|^2 dt = \int_0^1 |f(t+n)|^2 dt = \int_0^1 |\mu(n)f(t)|^2 dt = \int_0^1 |f(t)|^2 dt.$$

Summing over $n \in \mathbb{Z}$ gives $\|f\|^2 = \sum_n \int_0^1 |f|^2$. This is finite only if $\int_0^1 |f|^2 = 0$, and then $\|f\| = 0$, a contradiction. An irreducible closed invariant subspace would carry an irreducible representation with central character 0. By the classification above it would be a line. ∎

The following table compares central character $\lambda \neq 0$ with central character 0.

	central character $\lambda \neq 0$	central character 0
irreducibles	one class, Π_λ	$\chi_{\alpha,\beta}$, $\alpha,\beta \in \mathbb{R}$
dimension	infinite	one
the $L^2(\mathbb{R})$ model	Π_λ itself, irreducible	Π_0 has no irreducible piece
Gaussian coefficient	decays on $H/Z(H)$	constant in b

Self-checks

1. Compute gg' and $g'g$ for $g = (1,2,3)$ and $g' = (4,5,6)$, and check the commutation formula (18.2).
   Answer. $gg' = (5, 7, 3+6+5) = (5,7,14)$ and $g'g = (5,7,9+8) = (5,7,17)$. Here $ab' - a'b = 5 - 8 = -3$, and $(0,0,-3)(5,7,17) = (5,7,14)$.
2. Why are Π_2 and Π_3 inequivalent?
   Answer. An intertwiner T satisfies $Te^{2ic} = e^{3ic}T$ for all c. At $c = \pi$ this reads $T = -T$, so $T = 0$.

3. Compute $\langle \varphi, \Pi_1(2,2,2)\varphi\rangle$ and $\langle \varphi, \Pi_0(0,b,0)\varphi\rangle$.
   Answer. By the Gaussian coefficient (18.6), $c - ab/2 = 0$ and $(a^2+b^2)/4 = 2$, so the first is e^{-2}. The second is 1 for every b.

Summary

The Heisenberg group has centre $Z(H) \cong \mathbb{R}$, and every commutator lies in it. Translations and modulations on $L^2(\mathbb{R})$ form the Schrödinger representation Π_λ, which satisfies the Weyl relations. The Stone–von Neumann theorem, cited, says that for $\lambda \neq 0$ every strongly continuous unitary representation on a nonzero separable space with central character λ is a Hilbert direct sum of copies of Π_λ, up to unitary equivalence. At $\lambda = 0$ the irreducibles are the characters $\chi_{\alpha,\beta}$, and the translation representation Π_0 contains none of them.

Reading

- Kirillov, *Lectures on the Orbit Method* (2004), Chapter 2: the Heisenberg group, its irreducible unitary representations and the Stone–von Neumann theorem.
- Kirillov, *Lectures on the Orbit Method*, Appendix V: background on unitary representations, including Schur's lemma and induced representations.
- Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 1: the Heisenberg group among the examples of matrix Lie groups.